COINPICKS · RESEARCH & EDUCATION

Bitcoin Technical Analysis
Quant Pillar Report

The Tape Itself — Measured, Scored, and Mapped to the Pillar Standard

Machine-computed from 16 years of daily price data through a 7-indicator scoring stack, halving-cycle analogs, and a power-law valuation model. Every number is reproducible from the scripts in the BTC-TA folder. ***These numbers are computed, not judged, so they carry no draft/committed status of their own — they enter the master blend directly. This cycle's overall qualitative score set (2026-08-31_draft) is a light daily catch-up, AUTO-STAMPED: three qualitative pillars (CLARITY Act, Tokenization RWA, Fed & Macro) held flat, and Iran War Risk re-rated down on a resumed direct US strike on Iran plus an Iranian retaliatory strike on Jordan — see that pillar report. This quant pillar's own output is unaffected by qualitative status and enters the master blend the same way regardless.***

COINPICKS QUANT PILLAR REPORT · THE QUANTITATIVE SLEEVE OF MY THESIS

Data through August 31, 2026 (BTC $78,447.16) · Pipeline re-run & verified August 31, 2026 · Not Financial Advice

The Scores — August 31 Model Output

Computed August 31, 2026 (previous computed run August 29, 2026, a full weekly refresh) — same standard as every qualitative pillar. This cycle's mapping holds flat at the front end and firms slightly further out as the cycle-month clock advances: 1mo 54→54 (flat), 3mo 46→46 (flat), 6mo 56→59, 1yr 67→68, 3yr 69→69 (flat) — the weekly bar that closed above the 200-week MA on Aug 23 remains confirmed. This cycle's overall qualitative score set (2026-08-31_draft) is a light daily catch-up, AUTO-STAMPED (Iran War Risk re-rated down on a resumed direct US strike on Iran; the other three qualitative pillars held flat); this quant pillar is computed fresh every run regardless and carries no draft/committed status of its own.

WINDOW	MODEL P	QUALITY	IMPACT	SIGNAL	CALL
1 month	**54**	0.75	8	**+0.48**	Neutral, tilting bullish — the weekly close above the 200-week MA remains confirmed (weekly composite +34.2, Slight bullish, 6-of-7); the 1mo cycle-analog holds 1-of-3 positive, median −1.0%
3 months	**46**	0.70	8	**−0.45**	Neutral — still the softest medium window, 1-of-3 analog positive, median −10.7%
6 months	**59**	0.55	8	**+0.79**	Mild Bull — tilting up further as the 6mo analog turns unanimous, 3-of-3 positive, median +46.1%
1 year	**68**	0.60	8	**+1.73**	Bull — the cycle turn; every prior cycle at this exact month was higher a year later (3-of-3), median +77.9%
3 years	**69**	0.50	8	**+1.52**	Bull — value + cycle (36-month analog 3-of-3 positive, median +1235.7%)

Impact: 8 (the tape aggregates every driver, but adds no causal force of its own) · **Formula:** Signal = ((P − 50) ÷ 50) × Quality × Impact · **Honesty cap:** P = 50 + 25 × (window score ÷ 100) — pure TA never claims odds more than 25 points from a coin flip, because its measured out-of-sample edge is modest. Math machine-verified.

THE THREE INPUTS, IN PLAIN ENGLISH

P — the odds the tape is *good for Bitcoin* by then. Near-term still genuinely bullish, not just repairing (weekly composite +34.2/100, Slight bullish, agreement 6-of-7 — the weekly bar that closed above the 200-week MA on Aug 23 at $77,716.48 remains a confirmed completed-bar signal); long-term still high (at this exact point in prior cycles, the next 12 months were positive 3-for-3, median +77.9%, and price sits 44% below its power-law fair value).

Quality — how trustworthy the evidence is. Highest near-term (842 weekly bars of history feeding the indicator stack), lowest at 3 years (only 3 completed cycles to learn from, and this cycle is already breaking the old pattern).

Impact — 8 out of 10, now the uniform value shared by every pillar; the TA pillar carries an equal 20% weight in the master, same as each qualitative pillar. Price is the thing itself — but a mirror, not an engine.

The Claim

The thesis, and where the numbers differ from the mood

The crowd: −50% from the all-time high just weeks ago, a lower low on June 30, a weekly close under the 200-week average as recently as June 28 — "the cycle is dead, get out." (Price has since closed at $78,447.16, up **+1.0% versus Aug 29's $77,666.16** and +34.0% off the June 30 low, even as the Iran War Risk pillar re-rated down sharply this cycle on a resumed direct US strike on Iran and an Iranian retaliatory strike on Jordan; the tape moving on its own schedule is itself information: this move is coming from price action, not from the qualitative pillars.)

The model: the short-term tape remains genuinely confirmed, not just repaired — the weekly composite holds Slight bullish (+34.2, agreement 6-of-7) and the quarterly composite holds Slight bullish (+25.3, agreement 4-of-7); the monthly composite is essentially flat (+1.5, Neutral) with agreement narrow at 1-of-7 as its sign stays muddier near zero. The weekly bar dated 2026-08-23 that closed above the 200-week MA ($77,716.48) remains a confirmed completed-bar signal — Repair Signal #1 stays confirmed, not a lean. The 1-month window holds Neutral-tilting-bullish and flat (P 54 versus Aug 29) on that confirmation; 3 months holds Neutral (P 46, flat). The 6-month window firms further to Mild Bull (P 59) as the analog turns unanimous (3-of-3 positive, median +46.1%). The long read holds its direction (1yr 67→68, 3yr flat at 69): every prior cycle at this exact month was higher 12 months later (median now +77.9%), price is 44% below fair value ($138,979), and the froth that precedes real crashes remains absent.

The edge: the crowd reads one clock (the tape) — and the tape has held "confirmed bullish" at the near end. The model still reads three — tape (confirmed at the weekly level), cycle clock (bottoming window: prior cycles bottomed at months 26–30; we're at month 28.38, past the average-bottom date), and valuation (44% below the power-law fair line, with price sitting well above the −1σ "cheap" band). All three clocks point the same direction on the medium-to-long horizon — the near-term clock has stayed caught up to the other two.

The Facts — The Tape

*Computed August 31, 2026 from Bitstamp daily data (BTC $78,447.16, daily close as of August 31). Current weekly/monthly/quarterly bars are still forming — scores use them as-is. Quantified: on completed bars only, the weekly composite holds at **+31.2** (confirmed since the weekly bar dated 2026-08-23 closed above the 200-week MA); monthly −32.6 and quarterly +7.7 are unchanged, since their current bars (dated 2026-09-06, 2026-08-31, and 2026-09-30 respectively) are still forming.*

- **Weekly composite +34.2/100 (Slight bullish), agreement 6-of-7 indicators.** Price is above the 20-week ($70,076, +11.9%) and well above the **200-week average ($64,860, +20.9%)**; the gap to the 50-week has narrowed further ($80,307, −2.3%). RSI 57.3; MACD line still negative (−1,578) with histogram positive (+2,324) and growing — momentum repair continuing; %B 0.79. Price ($78,447.16) remains above the falling weekly resistance trendline ($76,216) that it broke above earlier this cycle.
- ★ **Repair signal #1 (a weekly close above the 200-week MA) remains CONFIRMED.** The weekly bar dated 2026-08-23 closed at $77,716.48, +20.9% above its 200-week average ($64,286.55) — a genuine completed-bar close above the line, not a still-forming daily read. The completed-bar weekly composite holds at **+31.2** since that close — the clearest single confirmation this pillar has logged this cycle.
- **Monthly composite +1.5/100 (Neutral), agreement narrow at 1-of-7** — essentially flat, up marginally from Aug 29's +0.1; the narrow agreement count reflects the composite sitting very close to zero. Below the 20-month average ($88,215, −11.1%), above the 50-month ($61,049, +28.5%); RSI 51.0; monthly MACD line +194 with histogram −4,782 — still negative; %B 0.36. The completed-bar monthly composite is −32.6, unchanged — this month's marginal improvement is still concentrated in the still-forming current bar (dated 2026-08-31).
- **Quarterly composite +25.3/100 (Slight bullish), agreement 4-of-7** — essentially holding from Aug 29's +24.5. Comfortably above the 20-quarter average ($58,144, +34.9%) and the 50-quarter ($28,734, +173.0%); RSI 57.3, MACD line +16,221 with histogram −438. Completed-bar quarterly composite is +7.7, unchanged (current bar dated 2026-09-30, still forming).
- **Levels that matter:** price ($78,447.16) sits between the old $74,434 resistance-turned-support (−5.1%, 1 touch) and fresh overhead resistance at $81,685 (+4.1%, 2x touch) — the nearest test on the board — then $89,164 (+13.7%, 1x) and $98,090 (+25.0%, 2x). Further below, $59,930 (−23.6%, 1x) and $57,735 (−26.4%, 1x). The falling weekly support trendline sits at $43,728; the falling resistance trendline, at $76,216, remains broken to the upside.
- ★ **June 30 remains the cycle low: $58,526 close** — the flush tagged the Fib golden pocket ($54,257–$57,802 = 61.8–65% retrace of the $15,479 → $126,272 run) and has since built a **+34.0% bounce to $78,447.16**. **$58,526 is still the line in the sand** — a decisive break back below it would undo this entire recovery leg.

Weekly tape, scored +34.2/100 (Slight bullish) with agreement 6 of 7 indicators: the weekly close above the 200-week MA remains confirmed on a completed bar, price holds above the falling resistance trendline, MACD histogram remains positive and growing. Machine-annotated from the same data that feeds the scores.

The Facts — The Cycle Clock & The Value Floor

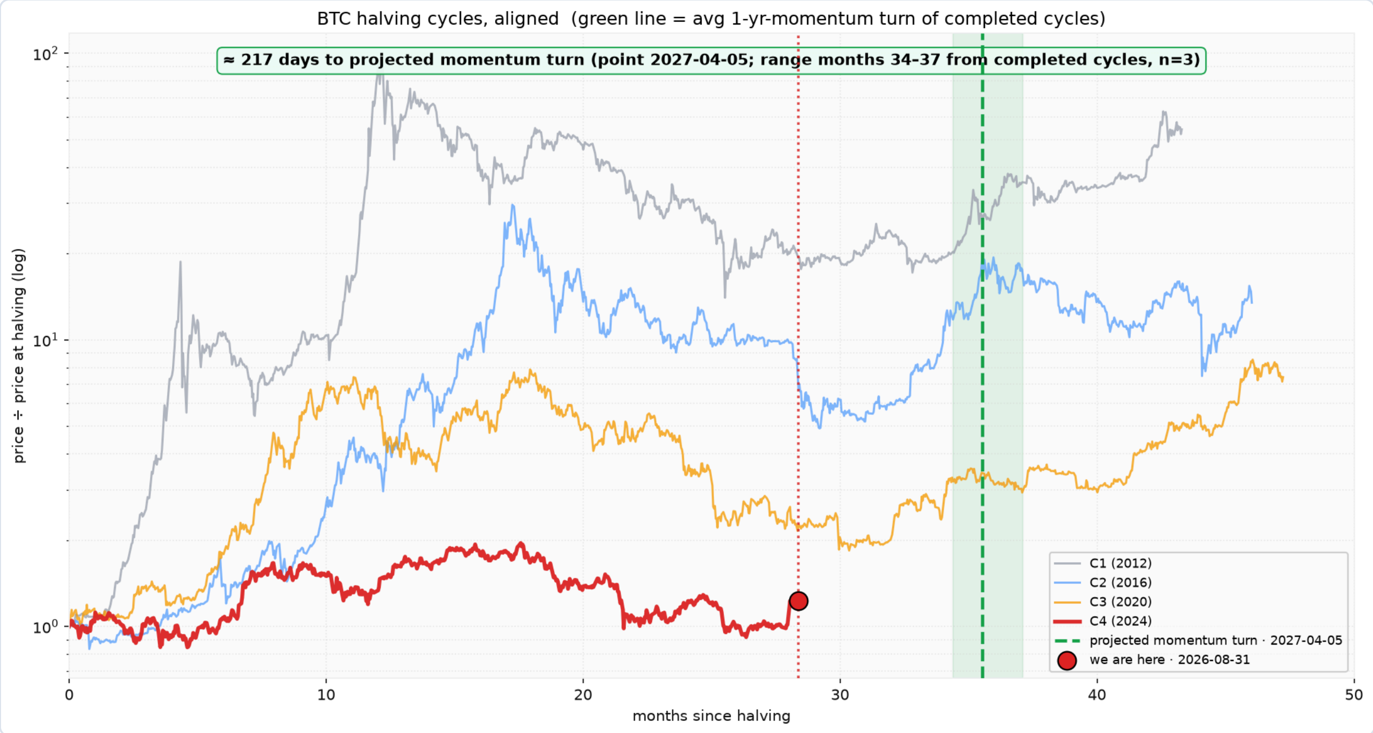
The two slower clocks. Same data, longer lenses.

- **The halving clock says "bottoming window."** The three completed cycles topped 12–18 months after their halving (avg 15.8) and bottomed at months 26–30 (avg 28.4). This cycle topped at month 18 (Oct 6, 2025 — $124,728) and printed its low-so-far at month 26 (June 30, $58,526, −53% from the top). **We are at month 28.38 — past the average-bottom date (~Aug 31, 2026), essentially today.**
- ★ **The cycle analogs are the spine of the long-window scores — and firmed further this cycle, with the 6-month window turning unanimous.** Standing at this exact cycle-month in C1/C2/C3: next 1 month positive **1-of-3** (median −1.0%) · next 3 months positive **1-of-3** (median −10.7%) · next 6 months positive **3-of-3** (median +46.1%, agreement firmed from 2-of-3) · next 12 months positive **3-of-3** (median +77.9%, up sharply from +47.0%) · next 36 months positive **3-of-3** (median +1235.7%, up from +986.0%). *Fragility disclosure: the medians and agreement counts moved as the cycle-month clock advanced from 28.32 to 28.38 — treat the analog levels as soft, the direction as the signal. And know what the 36-month number is: from month ~28 that window reaches month ~64 — it contains the NEXT halving's rally.*
- **1-year momentum turned positive at months 34–37 in every completed cycle (avg 35.5)** → still projects to **~April 2027** (80% window Jan–Jul 2027) for this cycle.
- **The power-law floor:** a log-log fit over the full price series puts fair value at **$138,979** — price is **−44%** versus fair. Fit-window honesty: starting the fit in 2011, 2012, 2013, or 2015 instead moves fair value to $128,822 / $133,781 / $117,230 / $144,104 respectively (fair value ranges roughly $117K–$144K depending on fit window) — the level is soft, but the **"well below fair" verdict holds across every window tested**. Price ($78,447.16) sits

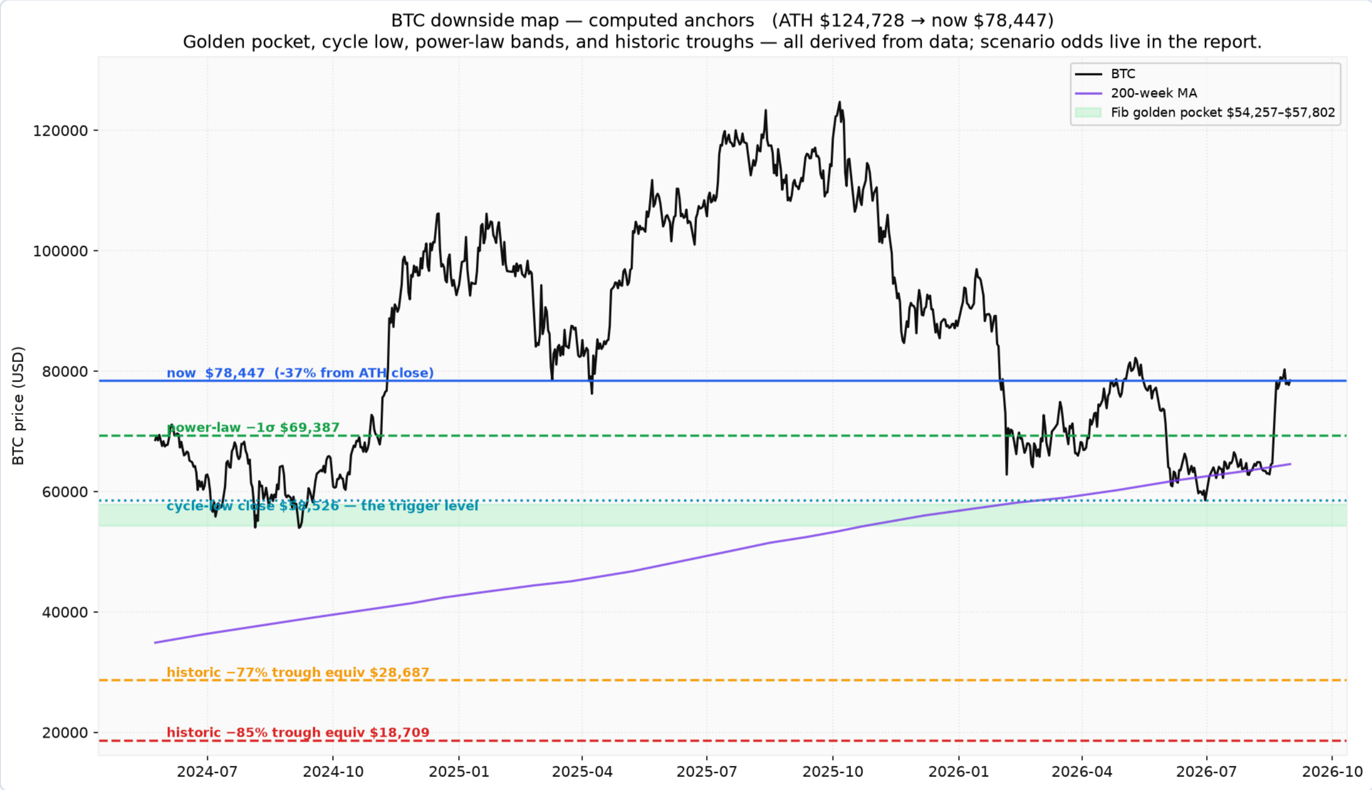
well above the **−1σ band ($69,387)** — out of the "cheap" zone (the −2σ band sits at $34,642, far below). Every halving-era cycle trough was −77% to −85% from ATH ($18.7K–$28.7K equivalent today) — the pre-halving 2011 bear went ~−93% (~$8.7K equivalent), excluded from the range but in the data; this cycle's low-so-far was −53% from the ATH.

- **No froth in the system:** the altcoin-froth index reads z = −0.87 — the opposite of the euphoric signature that preceded prior deep crashes. Validated on 16 years of data: high-froth days led to −39% average 12-month drawdowns vs −23% for low-froth days (gap significant, 95% CI excludes zero). *Data lags ~6 weeks (through May 23, unchanged) — used as a tilt, not a score input.*

★ **What we rejected:** a 5-feature logistic crash model was tested leave-one-cycle-out and **failed** (AUC 0.44 vs 0.58 for froth alone; 0.70 = useful). It is not in this report's numbers. The scores rest only on what survived out-of-sample testing.



All four halving cycles aligned (log scale, price ÷ price-at-halving). Triangles mark where 1-year momentum turned positive in each prior cycle — months 34–37, projecting to ~April 2027 for this one. The red dot is today: month 28.38, past the historical bottoming window.



The downside map, updated after the June 30 flush: the projected $54K–$61K bottom zone was hit ($58,526 close / $57,735 intraday, inside the golden pocket's doorstep) and bounced. Odds re-weighted accordingly: ~40% the low is in, ~35% one more flush to $54–58K, ~17% deeper bear, ~8% tail.

The Prize

Illustrative only — not price targets, not advice

From $78,447.16: the 20-week average (~$70.1K) is well behind us — price sits **+11.9%** above it; the 50-week average ($80,307) is back overhead as the immediate test, now just **+2.4%** away, while the falling weekly trendline resistance ($76,216) remains broken to the upside. A retest of the $124,728 ATH is **+59.0%**; the power-law fair line is **+77.2%**. The median 12-month analog from this cycle-month is now **+77.9%** (still shy of a full retest of the prior ATH, but a materially bigger read than last cycle's +47.0%); the median 36-month analog is +1235.7% — **discount that one hard**, it leans on regimes (2017 mania, post-COVID ZIRP) that don't repeat on demand.

The Prize doesn't feed the math — it tells you what the lean is worth acting on. Here it says: the near-term signal remains confirmed, not just repairing, and the medium-term analog picture has firmed further alongside it — the near quarter is still the softest window (P 46), but the 6-month-and-beyond case looks less lonely than it did last cycle, with the 6-month analog now unanimous (3-of-3).

The Other Side

The strongest honest case the model is wrong

- **n = 3.** Every long-window claim rests on three completed cycles. Three coin flips landing heads is not a law of nature.
- **This cycle already broke the pattern.** A −53% drawdown at month 26 vs −77/−85% in prior cycles could mean dampened cycles (bullish read) — or it could mean *the bottom isn't in*: a historic-depth trough from this ATH is $18.7K–$28.7K, far below every level in this report.

- **The ETF era changed the machine.** Prior cycles had no spot-ETF flow channel, no CME basis trade at this scale, no corporate treasuries. Analogs may simply not transfer.
- **The power law is a curve fit, not a promise.** R^2 0.96 in-sample says nothing about a structural break — a fit like this "said cheap" through most of 2022's decline too. It bounds scenarios; it doesn't time anything.
- **The tape's edge is small by construction.** The honesty cap exists because measured predictive validity is modest (froth AUC 0.58). If you removed the cap, this report would be louder — and less true.

The Triggers

Price and dated events — some prove the read right, some prove it wrong

WHEN / LEVEL	EVENT	MEANS
$58,526	Daily close below the June 30 low	Flush scenario live — golden pocket $54.3–57.8K is the target; re-rate 1/3-month down
~$64,286	Weekly close above the 200-week MA	**CONFIRMED** — the weekly bar dated 2026-08-23 closed at $77,716.48, +20.9% above this level; completed-bar weekly composite jumped −29.5 → +27.6 accordingly
~$80,307	Weekly close above the 50-week MA	Price sits −2.4% below it — the next real test; a confirmed weekly close above it would extend trend repair further — re-rate 1/3-month up if it holds
~Aug 31, 2026	Historical average-bottom month (28.4) passes	Arriving essentially today with price well off the June 30 low (+34.0%); a higher low + improving monthly momentum after this date = basing confirmed
Jan–Jul 2027	Projected 1-yr momentum turn window (point: ~Apr 2027)	Arriving without new lows = cycle thesis on track; re-rate 1-year up
$50,000	Weekly close below	−60% from ATH; the dampened-cycle story is failing — see the flip line

The flip line: a decisive weekly close below **$50,000** breaks the base case and the golden pocket both — treat the dampened-cycle thesis as broken and put the historic trough zone ($18.7K–$28.7K) back on the table. Above it, a bearish quarter is the plan, not a surprise.

My Decision

QUALITATIVE SET: COMMITTED (LIGHT DAILY CATCH-UP, AUTO-STAMPED) · THIS PILLAR: COMPUTED AUGUST 31, 2026 (NO DRAFT/COMMITTED STATUS OF ITS OWN)

Computed P: 54 / 46 / 59 / 68 / 69 · Quality: 0.75 / 0.70 / 0.55 / 0.60 / 0.50 · Impact: 8

This is the quantitative sleeve of the thesis — the same five windows, same formula, same table as every qualitative pillar, but computed instead of judged. The near-term read remains genuinely confirmed this cycle: the weekly bar that closed above the 200-week MA on Aug 23 stays a confirmed completed-bar signal, and the completed-bar weekly composite holds at +31.2 — the 1-month window holds flat at Neutral-tilting-bullish (P 54). The 6-month window firms further to Mild Bull (P 59) as the cycle-analog for that horizon turned unanimous (3-of-3 positive, median +46.1%). The 3-month window holds flat (P 46) as the softest window in the set. The long read holds its direction (1yr 67→68, 3yr flat at 69) as the 1-year analog median jumped to +77.9% from +47.0% as the cycle-month clock advanced — the direction, not the level, is the signal.

These numbers enter the master blend exactly as computed — this pillar has no override/re-stamp step of its own. Unlike the four qualitative pillars, this pillar carries no draft/committed status of its own: this cycle's overall qualitative score set (2026-08-31_draft) is a light daily catch-up, AUTO-STAMPED per delegated daily authority (Iran War Risk re-rated down on a resumed direct US strike on Iran; the other three qualitative pillars held flat). This pillar's own computed output is unaffected by that process and enters the master blend the same way it does every cycle.

What Changed Since Last Check

Pipeline re-run August 31, 2026 (previous computed run August 29, 2026, a full weekly refresh).

- **Price: $77,666.16 (Aug 29) → $78,447.16 (Aug 31), +1.0% net.** June 30's $58,526 remains the cycle low; price is now +34.0% off it.
- ★ **Repair signal #1 remains CONFIRMED: the weekly bar dated 2026-08-23 closed at $77,716.48, +20.9% above the 200-week average ($64,286.55).** This is a genuine completed-bar close, not a still-forming daily read — the strongest single confirmation this pillar has logged this cycle.
- **Composites: the completed-bar weekly composite holds at +31.2** (up from +27.6 as the current weekly bar, dated 2026-09-06, extends the confirmed trend) — confirmed since the Aug 23 weekly close above the 200-week MA. Monthly and quarterly completed-bar composites held flat (−32.6, +7.7) since their current bars are still forming. On the current (forming) bars, versus Aug 29: weekly +31.4→+34.2 (agreement firmed 5-of-7→6-of-7, still Slight bullish), monthly +0.1→+1.5 (agreement narrow at 1-of-7, still essentially flat), quarterly +24.5→+25.3 (agreement 4-of-7, still Slight bullish).
- **Model P vs Aug 29: 54/46/56/67/69 → 54/46/59/68/69.** 1mo flat; 3mo flat, still the softest window; 6mo +3 as the cycle-analog for that window turned unanimous; 1yr +1; 3yr flat.
- **Cycle analogs firmed further this cycle, most notably at 6mo and 1yr:** agreement counts moved to 1/3, 1/3, 3/3, 3/3, 3/3 (the 6-month window's agreement firmed from 2/3 to 3/3) as the cycle-month clock advanced from 28.32 (Aug 29) to 28.38 (Aug 31), with medians moving (1mo −0.6%→−1.0%, 3mo −10.5%→−10.7%, 6mo +47.5%→+46.1%, 1yr +47.0%→+77.9%, 3yr +986.0%→+1235.7%) — flagged per this report's standing n=3 fragility disclosure, not treated as new structural information.
- **Power-law fair value: $138,805 (Aug 29) → $138,979 (Aug 31)** (price −44% → −44% vs. fair) — normal drift as new daily bars extend the fit.
- **Data refreshed:** Bitstamp daily through August 31, 2026 (5,889 days on the master series); Coin Metrics altcoin set still through May 23 (community data lags ~6 weeks — froth is a tilt, not a score input).

- **Context:** this cycle's overall qualitative score set (2026-08-31_draft) is a light daily catch-up, AUTO-STAMPED per delegated daily authority — Iran War Risk re-rated down on a resumed direct US strike on Iran and an Iranian retaliatory strike on Jordan; the other three qualitative pillars held flat. This pillar's own computed output is independent of that process and enters the master blend the same way regardless of the qualitative set's status.

Method & Sources

Everything is reproducible — the scripts and data live in the BTC-TA folder. Run order below.

- **Price data:** Bitstamp OHLC API (daily, 2011–2026, no key) via fetch_btc_history.py · Coin Metrics community reference price (2010–2011 close-only) via build_master_history.py — 5,835 days total, each row tagged by source.
- **Indicator stack:** indicators.py — SMA 20/50/200, EMA 12/26, RSI-14 (Wilder), MACD 12/26/9, Bollinger 20/2, OBV — on weekly (W-SUN), monthly, quarterly bars. levels.py — swing-cluster S/R + least-squares trendlines.
- **Scoring:** score.py — each indicator → conviction in [−100, +100] (bounded inputs mapped linearly, unbounded squashed with tanh); composite = fixed weights (MA .20, RSI .15, MACD .15, BB .15, S/R .15, OBV .10, TL .10); agreement = indicators sharing the composite's sign.
- **Cycle & value:** cycle_analysis.py — halving-aligned tops/bottoms/momentum turns, projections from C1–C3 · downside.py / downside_map.py — power-law bands (R^2 0.96) + historical troughs + 6-method bottom-zone synthesis · test_altcoin_froth.py — froth quartiles vs forward drawdowns, block-bootstrap CI.
- **The mapping (new):** pillar_mapping.py — window scores from tape composites + cycle analogs + power-law value; P = 50 + 25×(S/100); Quality rubric per window; writes data/pillar_scores.json for the master blend.
- **Run order:** fetch_btc_history → build_master_history → fetch_altcoin_data → indicators → levels → score → charts → cycle_analysis → downside → downside_map → pillar_mapping.
- **Honesty notes:** current weekly/monthly/quarterly bars are partial as of Aug 31 (completed-bar composites: weekly +31.2, confirmed by the Aug 23 weekly close above the 200-week MA / monthly −32.6 / quarterly +7.7 — monthly and quarterly unchanged since their current bars are still forming) · the multi-signal crash model failed out-of-sample (AUC 0.44) and is excluded · the AUC figures are the pipeline's own attestation · froth data lags ~6 weeks · n=3 on all cycle claims, and analog windows can cross into the next cycle — the 6-month analog remains fragile on pure alignment drift, printed in the Facts · the 2024 halving is dated Apr 19 (US ET; block 840,000 hit Apr 20 00:09 UTC — 0.03 cycle-month effect) · S/R levels are algorithmic swing-clusters with 1–2 touches each · the bottom-odds split (40/35/17/8) is a judgment synthesis, not a model output.